

National University of Singapore
School of Computing
CS1101S: Programming Methodology
Semester I, 2026/2027

S5
Box-and-Pointer Diagrams

For this sheet, draw the box-and-pointer diagrams by hand (e.g. using pen and paper) and bring the drawings to your Studio.

Problems:

1. Draw box-and-pointer for the values of the following expressions. Also give box and llist notation.

```
llist(llist(1, 2, llist(3)), llist(4, 5), pair(6, 7))
```



```
pair(1, llist(2, 3, pair(4, None)))
```



```
pair(1, pair(2, llist(3, llist(4, 5))))
```



2. Examine the following (incorrect) reverse function:

```
def reverse(llst):  
    return None if is_none(llst) else pair(reverse(tail(llst)),  
                                             head(llst))
```



Draw the box-and-pointer diagram, box notation and llist notation for the result returned by the following function call:

```
reverse(llist(1, 2, 3, 4))
```



3. Write expressions using `llst`, `head` and `tail` that will return 1 when the `llst` is bound to the following values:

```
llist(7, llist(6, 5, 4), 3, llist(2, 1))
```



```
llist(llist(7), llist(6, 5, 4), llist(3, 2), 1)
```



```
l1ist(7, l1ist(6), l1ist(5, l1ist(4)), l1ist(3, l1ist(2, l1ist(1))))
```



```
l1ist(7, l1ist(l1ist(6, 5), l1ist(4), 3, 2), l1ist(l1ist(1)))
```

